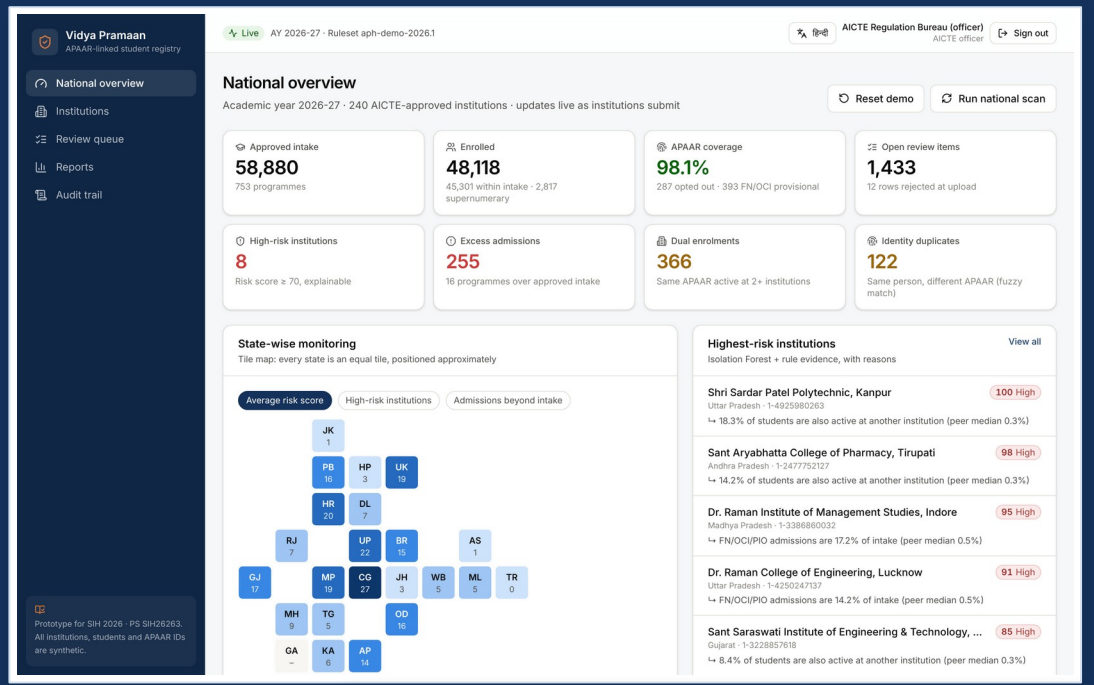
SMART INDIA HACKATHON 2026 · IDEA SUBMISSION

Vidya Pramaan

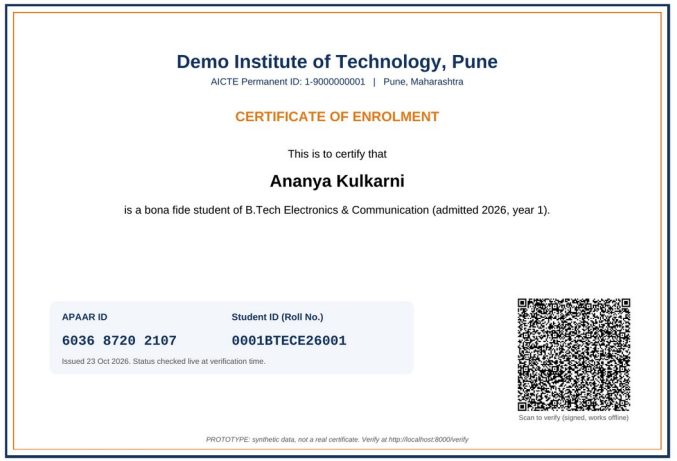
APAAR ID-based Student Data Management & Verification Platform for AICTE-Approved Institutions

One student. One APAAR ID. One verifiable record.

Problem Statement ID **SIH26263**	Problem Statement Title **APAAR ID-based student data platform**
Theme **Smart Automation**	PS Category **Software**
Team ID **(as allotted on SIH portal)**	Team Name **AI Riders**



Working prototype: national dashboard and a certificate carrying APAAR ID + student ID + signed QR



PROPOSED SOLUTION

An APAAR-keyed student register, validated before storage

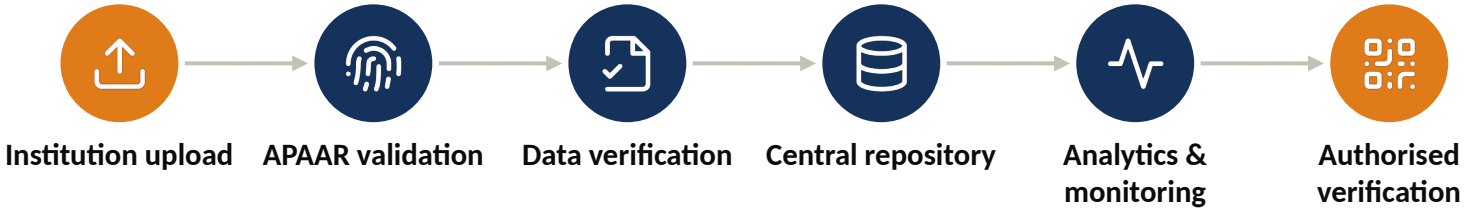
The gap today

- Only PGDM institutions send student-level data; the rest report totals at approval time
- Excess admissions and dual enrolment surface late, if at all
- Employers and regulators cannot verify a student's record at source

What we built

A portal and API where each AICTE-approved institution submits student data after admissions. Every row is checked against APAAR, the approved intake and every other institution before it is stored. AICTE then monitors it live, and employers verify it only with the student's consent.

Inclusive by design: students who opt out of APAAR, and FN / OCI students, get a provisional ID. They are noted, never rejected.



Explainable validation

25 rule codes (APR, DUP, INT, CON), each with evidence and the ruleset version

Live seat ledger

Admission #61 in a 60-seat programme is stopped at upload; supernumerary caps per APH

Cross-institution dedup

Same APAAR at two colleges, and the same person under a different APAAR (fuzzy match)

Explainable risk score

Isolation Forest + rule evidence ranks institutions, with plain-language reasons

Consent-first verification

Student approves by OTP; the verifier sees only the fields requested

Signed records, sealed audit

W3C credential + QR on certificates; hash-chained log exposes tampering

TECHNICAL APPROACH

Modular and API-first: one codebase from demo to production

Channels

- Institution portal (Excel / CSV upload)
- ERP & university REST API
- Student portal (consent, credentials)
- Employer / regulator verifier

↓

Gateway

- **Role-based access with institution scoping, session expiry → production: AICTE SSO (OIDC + MFA), rate limits**

↓

Services

- Validation engine (rules as code)
- APAAR registry adapter mock → API Setu
- National scan + ML risk score
- Consent manager + access log
- Credential issuer Ed25519 · QR

↓

Data

- SQLite now → PostgreSQL, encrypted PII
- Hash-chained audit log
- Live dashboards (SSE) & reports (CSV / XLSX)

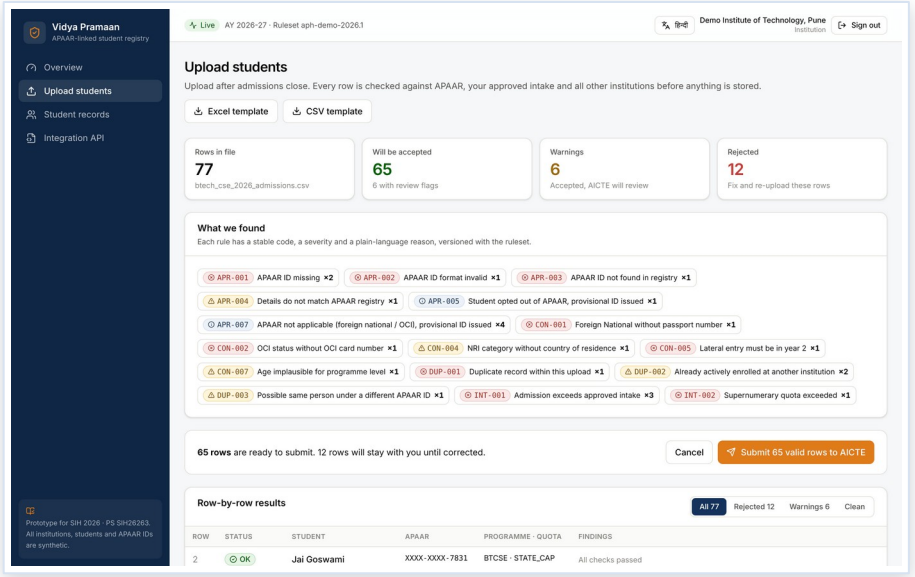
Everything shown runs in the prototype today; text after → is the production upgrade.

How a submission flows

Upload → 25 rule checks per row (57 ms for 77 rows) → valid rows committed, rejected rows returned with reasons → national re-scan across 48,053 enrolments (1.03 s) → risk scores refreshed → AICTE dashboard updates live

Stack

- React · TypeScript · Tailwind
- FastAPI (Python)
- PostgreSQL (SQLite in demo)
- scikit-learn · RapidFuzz
- Ed25519 · W3C VC · QR
- OpenAPI 3 · REST + SSE



Institution upload: 77 rows, 65 accepted, 12 rejected with rule codes

Feasible today: measured on 48,000 synthetic student records

16/16
planted defect types caught at upload

181/181
dual enrolments found across colleges

60/60
same person, different APAAR found

41/42
top-ranked institutions were truly anomalous

0
false flags on 56 clean rows

1.03 s
national scan of 48,053 enrolments

Synthetic data with planted anomalies (docs/benchmark.json). Also covered by 16 automated API tests and a repeatable browser walkthrough of every role. Real data will be noisier; every flag goes to a human reviewer.

Viability

- Open-source stack, no licence cost; runs on NIC / MeghRaj cloud for data localisation
- Reuses APAAR, DigiLocker and NAD/ABC instead of creating a parallel silo
- Starts where data already exists: PGDM institutions report student-level data today

Risk / challenge	Mitigation
No public APAAR API for third parties	Registry adapter: mock now; DigiLocker / API Setu partner onboarding for production
Students without APAAR (opt-out, FN / OCI)	Provisional AICTE ID with an INFO note; never rejected
Privacy: DPDP Act 2023 & Rules 2025	Consent ledger, purpose-bound access, masked IDs, access log the student can see
False positives hurting institutions	Flags are review items; an officer decides; every decision is audited
Low digital maturity in small colleges	Excel template with drop-downs, plain-language errors, English / Hindi toggle, works on mobile
National scale	Stateless API, partitioned PostgreSQL, async bulk jobs, read replicas for dashboards

IMPACT & BENEFITS

Authentic data, lighter compliance, portable proof

AICTE & regulators

- Live admissions vs approved intake
- Excess and dual enrolment caught at submission
- Evidence for approval decisions

Institutions

- One upload replaces repeated returns
- Errors explained row by row before submission
- Data-quality score instead of surprise inspections

Students

- Lifelong record tied to APAAR
- Certificate with APAAR ID, student ID and QR
- Control over who sees what, and an access log

Employers & universities

- Instant QR check, offline signature
- Consent-based verification of specific facts
- Fewer fake-certificate cases

Policy makers

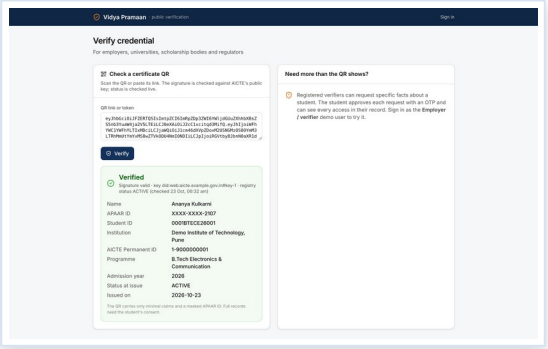
- State, gender, category and level analytics
- Intake utilisation to plan seat approvals
- Exports for NBA / NAAC / AISHE-style reports

Path to adoption

Phase 1 · 3 months
Pilot with PGDM institutions (already report student data) and one state

Phase 2 · 6 months
DigiLocker / API Setu APAAR integration; push verified records to NAD / ABC

Phase 3 · next AY
All AICTE institutions; certificate QR adopted by affiliating universities



Public QR check: signature valid, status live

What our research changed in the design

APAAR creation needs Aadhaar-based KYC
→ FN / OCI students may not have one, so they get a provisional ID (rule APR-007)

APAAR consent is voluntary
→ A student who opts out is recorded, not rejected (rule APR-005)

DPDP Rules 2025 phase in to 2027
→ Consent artefacts, purpose limitation and student-visible access logs built in from day one

AICTE's June 2026 circular: APAAR-linked records to NAD / ABC
→ Verified records flow to NAD / ABC rather than a parallel silo

APH sets supernumerary quotas (FN/OCI/PIO, TFW, lateral, EWS)
→ Caps live in configuration, versioned with the ruleset

References

1. SIH 2026 PS SIH26263, AICTE Regulation Bureau (sih.gov.in)
2. AICTE Approval Process Handbook 2024-27
3. AICTE circular, 23 Jun 2026: upload of APAAR-linked records to NAD/ABC
4. APAAR / Academic Bank of Credits (abc.gov.in), DigiLocker
5. API Setu: DigiLocker Requester & Issuer API specifications
6. Digital Personal Data Protection Act 2023; DPDP Rules 2025
7. W3C Verifiable Credentials Data Model; RFC 8032 (Ed25519)
8. Liu, Ting & Zhou (2008), Isolation Forest, IEEE ICDM
9. Fellegi & Sunter (1969), A Theory for Record Linkage, JASA

Prototype, code & tests
github.com/Pulkitkh/AI_Riders_
Synthetic data only